

Dhathri Desai Yarramaddu

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LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science graduate student with experience in machine learning, computer vision, and data engineering. Co-authored a published research paper on automated spinal stenosis classification using deep learning. Skilled in Python, Java, SQL, Apache Spark, Kafka, and Databricks. Strong foundation in software development and data-driven systems.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, MySQL

Data Engineering: Apache NiFi, Apache Spark, MongoDB, Hive, Apache Kafka, Azure Data Factory, Databricks

AI / ML: Machine Learning, Deep Learning, ResNet50V2, CNN, YOLOv5, Grad-CAM, Model Evaluation

Data & Visualization: Tableau, Jupyter Notebook, Google Colab, Streamlit

Tools: Git, Linux, Microsoft Office, Google Workspace

Core Strengths: Communication, Teamwork, Documentation, Organization, Adaptability

EXPERIENCE

Data Engineering Intern — India

JUN 2024 - NOV 2024

- Developed and optimized ETL pipelines using Apache NiFi and Apache Spark for large-scale financial data ingestion and processing.
- Enhanced pipeline performance by 15% through Spark optimization, parallel processing, and automated error handling and validation.
- Worked with MongoDB and Hive for structured and unstructured data, improving query performance by 30% using indexing, partitioning, and Parquet/ORC formats.
- Built real-time and batch workflows using Azure Data Factory, Databricks, and Apache Kafka for scalable low-latency analytics.

PROJECTS

Automated Detection and Classification of Spinal Stenosis

2024 - 2025

- Developed a ResNet50V2-based deep learning model for lumbar spinal stenosis classification from MRI images.
- Applied image preprocessing and feature extraction techniques to improve classification accuracy and model reliability.
- Used Grad-CAM for interpretability to highlight important regions in MRI images supporting model predictions.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix to validate diagnostic effectiveness.

AI-Driven Aircraft Maintenance System

2024

- Built a YOLOv5-based computer vision model to detect cracks in aircraft inspection images.
- Developed predictive models for aircraft battery health to support preventive maintenance planning.
- Created a visualization dashboard to display defect detection results and maintenance-related insights.
- Applied AI-based inspection methods to reduce manual effort and improve maintenance decision-making.

Superstore Sales Analysis and Visualization

2023

- Performed exploratory data analysis on retail sales data to identify revenue trends and business insights.
- Built dashboards to visualize regional sales performance, profitability, and category-level trends.

EDUCATION

DEPAUL UNIVERSITY, CHICAGO

2025 - 2027

- Master of Science in Computer Science

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI

2021 - 2025

- Bachelor of Technology in Electronics and Communication Engineering

RESEARCH PAPERS

ATLANTIS PRESS / ICARECS 2025 - AUTOMATED DETECTION AND CLASSIFICATION OF SPINAL STENOSIS **JUNE 2025**

- Co-authored a conference paper on AI-based spinal stenosis detection and classification using deep learning and medical imaging.
- Published in the Proceedings of the 2025 International Conference on Advanced Research in Electronics and Communication Systems.
- DOI: 10.2991/978-94-6463-754-0_47

CERTIFICATIONS

Cybersecurity Analyst Job Simulation — Tata Consultancy Services

Python Programming Bootcamp

Artificial Intelligence and Machine Learning Foundations

ACTIVITIES & LEADERSHIP

Directorate of Student Affairs, SRM Institute of Science and Technology

- Volunteered in organizing student activities and campus events.
- Assisted in logistics, coordination, and communication for student programs.